Machine Learning for Mechanical Engineers at CoEP Session 18: Titanic Capstone

Session 18: Titanic

Case Study: Titanic Disaster

The Disaster

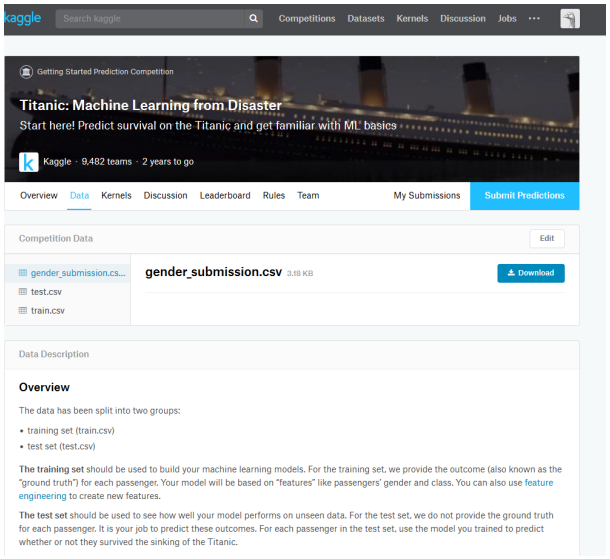
(Image Reference: https://en.wikipedia.org/wiki/Willy_Stower)

The Titanic competition

- The sinking of the Titanic is one of the most infamous shipwrecks in history.
- It sank after colliding with an iceberg, killing 1502 out of 2224 passengers and crew.
- One of the reasons for loss of life was that there were not enough lifeboats for the passengers and crew.
- Some groups of people were more likely to survive than others, such as women, children, and the upper-class.

Kaggle

Kaggle Competition



What is Kaggle?

- Kaggle is a site where people create algorithms and compete against machine learning practitioners around the world.
- Your algorithm wins the competition if it's the most accurate on a particular data set.
- Kaggle is a fun way to practice your machine learning skills.
- Indian counterpart is "Analytics Vidhya".
- Many competitions and good prizes.

The Titanic competition

- Kaggle has created a number of competitions designed for beginners.
- One of the most popular of these competitions is Titanic competition
- In this competition, we have a data set of different information about passengers onboard the Titanic, and we see if we can use that information to predict whether those people survived or not.

The Titanic competition

- Each Kaggle competition has two key data files that you will work with - a training set and a testing set.
- The training set contains data we can use to train our model.
- It has a number of feature columns which contain various descriptive data, as well as a column of the target values we are trying to predict: in this case, Survival.
- The testing set contains all of the same feature columns, but is missing the target value column. Additionally, the testing set usually has fewer observations (rows) than the training set.

Data

Data Dictionary

Below are the descriptions contained in that data dictionary

- PassengerID: A column added by Kaggle to identify each row
- Survived: Passenger survived or not (0=No, 1=Yes)
- Pclass: The class of the ticket (1=1st, 2=2nd, 3=3rd)
- Sex: The passenger's sex
- Age: The passenger's age in years
- SibSp: The number of siblings or spouses
- Parch: The number of parents or children
- Ticket: The passenger's ticket number
- Fare: The fare the passenger paid
- Cabin: The passenger's cabin number
- Embarked: The port where the passenger embarked (C=Cherbourg, Q=Queenstown, S=Southampton)

Load the data

```
import pandas as pd

test = pd.read_csv("titanic_test.csv")
train = pd.read_csv("titanic_train.csv")

print("Dimensions of train: {}".format(train.shape))
print("Dimensions of test: {}".format(test.shape))

Dimensions of train: (891, 12)
Dimensions of test: (418, 11)
```

Data Exploration

Exploring the data

Let's take a look at the first few rows of the train dataframe.

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35	0	0	373450	8.0500	NaN	S

```
train.head()
```

Exploring the data

Let's take a look at the last few rows of the train dataframe.

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
886	887	0	2	Montvila, Rev. Juozas	male	27	0	0	211536	13.00	NaN	S
887	888	1	1	Graham, Miss. Margaret Edith	female	19	0	0	112053	30.00	B42	S
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.45	NaN	S
889	890	1	1	Behr, Mr. Karl Howell	male	26	0	0	111369	30.00	C148	C
890	891	0	3	Dooley, Mr. Patrick	male	32	0	0	370376	7.75	NaN	Q

```
train.tail()
```

Column Data Types

df_train.dtypes

- Type 'object' is a string for pandas, which poses problems with machine learning algorithms.
- If we want to use these as features, we'll need to convert these to number representations.

```
PassengerId      int64
Survived         int64
Pclass           int64
Name            object
Sex             object
Age            float64
SibSp            int64
Parch            int64
Ticket          object
Fare           float64
Cabin           object
Embarked        object
```

Dataframe information

df_train.info()

- Age, Cabin, and Embarked are missing values.
- Cabin has too many missing values.

```
Data columns (total 12 columns):
PassengerId    891 non-null int64
Survived       891 non-null int64
Pclass         891 non-null int64
Name           891 non-null object
Sex            891 non-null object
Age            714 non-null float64
SibSp          891 non-null int64
Parch          891 non-null int64
Ticket         891 non-null object
Fare           891 non-null float64
Cabin          204 non-null object
Embarked       889 non-null object
```

Descriptive Statistics

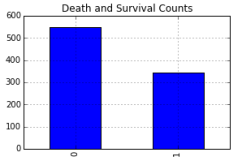
df_train.describe()

- Age, Cabin, and Embarked are missing values.
- Cabin has too many missing values.

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

Plot Features

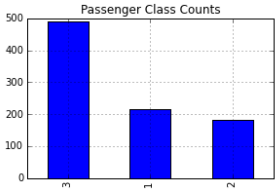
Plot death and survival counts




```
# Set up a grid of plots
fig = plt.figure(figsize=fizsize_with_subplots)
fig_dims = (3, 2)

plt.subplot2grid(fig_dims, (0, 0))
df_train['Survived'].value_counts().plot(kind='bar',
                                         title='Death and
    Survival Counts')
```

Plot Features

Plot Pclass counts




```
plt.subplot2grid(fig_dims, (0, 1))
df_train['Pclass'].value_counts().plot(kind='bar',
                                       title='Passenger
    Class Counts')
```

Plot Features

Plot Sex counts

```
plt.subplot2grid(fig_dims, (1, 0))
df_train['Sex'].value_counts().plot(kind='bar',
                                    title='Gender Counts')
                                    plt.xticks(rotation=0)

# Plot Embarked counts

plt.subplot2grid(fig_dims, (1, 1))
df_train['Embarked'].value_counts().plot(kind='bar',
                                         title='Ports of
    Embarkation Counts')

# Plot the Age histogram

plt.subplot2grid(fig_dims, (2, 0))
df_train['Age'].hist()
plt.title('Age Histogram')
```

Cross Tab

We'll determine which proportion of passengers survived based on their passenger class.. Generate a cross tab of Pclass and Survived:

Survived	0	1
Pclass		
1	80	136
2	97	87
3	372	119

```
pclass_xt = pd.crosstab(df_train['Pclass'], df_train['
    Survived'])
pclass_xt
```

Feature Engineering

Generate a mapping of Sex from a string to a number representation:

```
sexes = sorted(df_train['Sex'].unique())
genders_mapping = dict(zip(sexes, range(0, len(sexes) + 1)
    ))
genders_mapping

{'female': 0, 'male': 1}
```

Feature Engineering

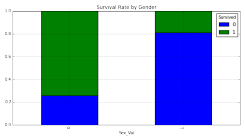
Transform Sex from a string to a number representation:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	Sex_Val
0	1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171	7.2500	NaN	S	1
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38	1	0	PC 17599	71.2833	C85	C	0
2	3	1	3	Heikkinen, Miss. Laina	female	26	0	0	STON/O2. 3101282	7.9250	NaN	S	0
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803	53.1000	C123	S	0
4	5	0	3	Allen, Mr. William Henry	male	35	0	0	373450	8.0500	NaN	S	1

```
df_train['Sex_Val'] = df_train['Sex'].map(genders_mapping)
    .astype(int)
df_train.head()
```

Exploring the data

Plot a normalized cross tab for Sex_Val and Survived:



The majority of females survived, but not males.

```
sex_val_xt = pd.crosstab(df_train['Sex_Val'], df_train['
    Survived'])
sex_val_xt_pct = sex_val_xt.div(sex_val_xt.sum(1).astype(
    float), axis=0)
sex_val_xt_pct.plot(kind='bar', stacked=True, title='
    Survival Rate by Gender')
```

Exploring the data

Any insights by looking at both Sex and Pclass? Count males and females in each Pclass:

```
passenger_classes = sorted(df_train['Pclass'].unique())

for p_class in passenger_classes:
    print('M: ', p_class, len(df_train[(df_train['Sex'] ==
      'male') &
                               (df_train['Pclass'] ==
     p_class)]))
    print('F: ', p_class, len(df_train[(df_train['Sex'] ==
      'female') &
                               (df_train['Pclass'] ==
     p_class)]))

M:  1 122
F:  1 94
M:  2 108
F:  2 76
M:  3 347
F:  3 144
```

Exploring the data

Plot survival rate by Sex:

```
# Plot survival rate by Sex
females_df = df_train[df_train['Sex'] == 'female']
females_xt = pd.crosstab(females_df['Pclass'], df_train['
    Survived'])
females_xt_pct = females_xt.div(females_xt.sum(1).astype(
    float), axis=0)
females_xt_pct.plot(kind='bar',  stacked=True,
                    title='Female Survival Rate by
    Passenger Class')
plt.xlabel('Passenger Class')
plt.ylabel('Survival Rate')
```

Exploring the data

Plot survival rate by Pclass:

The vast majority of females in First and Second class survived. Males in First class had the highest chance for survival.

```
# Plot survival rate by Pclass
males_df = df_train[df_train['Sex'] == 'male']
males_xt = pd.crosstab(males_df['Pclass'], df_train['
    Survived'])
males_xt_pct = males_xt.div(males_xt.sum(1).astype(float),
     axis=0)
males_xt_pct.plot(kind='bar',    stacked=True,
                  title='Male Survival Rate by Passenger
    Class')
plt.xlabel('Passenger Class')
plt.ylabel('Survival Rate')
```

Cleaning the data

The Embarked column might be an important feature but it is missing a couple data points which might pose a problem for machine learning algorithms:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	Sex_Val
61	62	1	1	Icard, Miss. Amelie	female	38	0	0	113572	80	B28	NaN	0
829	830	1	1	Stone, Mrs. George Nelson (Martha Evelyn)	female	62	0	0	113572	80	B28	NaN	0

```
df_train[df_train['Embarked'].isnull()]
```

Cleaning the data

Prepare to map Embarked from a string to a number representation:

```
# Get the unique values of Embarked
embarked_locs = sorted(df_train['Embarked'].unique())

embarked_locs_mapping = dict(zip(embarked_locs,
                                 range(0, len(
    embarked_locs) + 1)))
embarked_locs_mapping

{nan: 0, 'C': 1, 'Q': 2, 'S': 3}
```

Cleaning the data

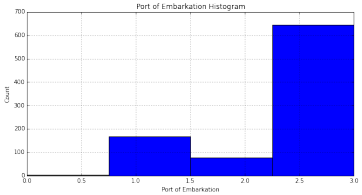
Transform Embarked from a string to a number representation to prepare it for machine learning algorithms:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	Sex_Val
0	1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171	7.2500	NaN	S	1
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38	1	0	PC 17599	71.2833	C85	C	0
2	3	1	3	Heikkinen, Miss. Laina	female	26	0	0	STON/O2. 3101282	7.9250	NaN	S	0
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803	53.1000	C123	S	0
4	5	0	3	Allen, Mr. William Henry	male	35	0	0	373450	8.0500	NaN	S	1

```
df_train['Embarked_Val'] = df_train['Embarked'] \
                               .map(embarked_locs_mapping)
    \
                               .astype(int)
df_train.head()
```

Cleaning the data

Plot the histogram for Embarked_Val:




```
df_train['Embarked_Val'].hist(bins=len(embarked_locs),
     range=(0, 3))
plt.title('Port of Embarkation Histogram')
plt.xlabel('Port of Embarkation')
plt.ylabel('Count')
plt.show()
```

Cleaning the data

Since the vast majority of passengers embarked in 'S': 3, we assign the missing values in Embarked to 'S':

```
if len(df_train[df_train['Embarked'].isnull()] > 0):
    df_train.replace({'Embarked_Val' :
                { embarked_locs_mapping[nan] :
     embarked_locs_mapping['S']
                }
            },
            inplace=True)
```

Cleaning the data

Verify we do not have any more NaNs for Embarked_Val:

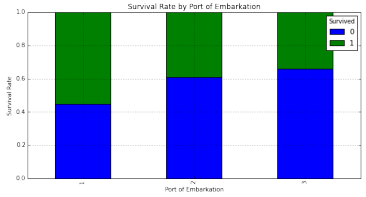
```
embarked_locs = sorted(df_train['Embarked_Val'].unique())
embarked_locs

array([1, 2, 3])
```

Exploring the data

Plot a normalized cross tab for Embarked_Val and Survived:

```
embarked_val_xt = pd.crosstab(df_train['Embarked_Val'],
     df_train['Survived'])
embarked_val_xt_pct = \
    embarked_val_xt.div(embarked_val_xt.sum(1).astype(
     float), axis=0)
embarked_val_xt_pct.plot(kind='bar', stacked=True)
plt.title('Survival Rate by Port of Embarkation')
plt.xlabel('Port of Embarkation')
plt.ylabel('Survival Rate')
```

Exploring the data



It appears those that embarked in location 'C': 1 had the highest rate of survival.

Feature Engineering

The Age column seems like an important feature–unfortunately it is missing many values. Filter to view missing Age values:

	Sex	Pclass	Age
5	male	3	NaN
17	male	2	NaN
19	female	3	NaN
26	male	3	NaN
28	female	3	NaN

```
df_train[df_train['Age'].isnull()][['Sex', 'Pclass', 'Age'
    ]].head()
```

Feature Engineering

Determine the Age typical for each passenger class by Sex_Val. We'll use the median instead of the mean because the Age histogram seems to be right skewed.

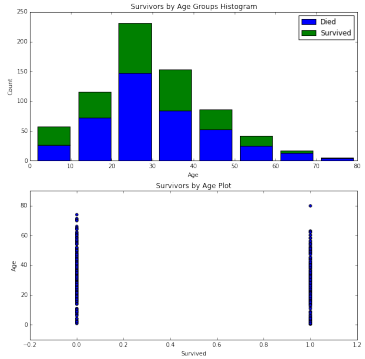
```
# To keep Age in tact, make a copy of it called AgeFill
# that we will use to fill in the missing ages:
df_train['AgeFill'] = df_train['Age']

df_train['AgeFill'] = df_train['AgeFill'] \
                      .groupby([df_train['Sex_Val'],
     df_train['Pclass']]) \
                      .apply(lambda x: x.fillna(x.median
     ()))
```

Feature Engineering

Plot a normalized cross tab for AgeFill and Survived:

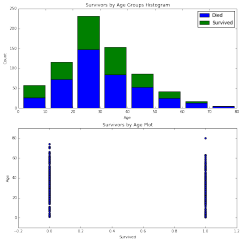
```
# Set up a grid of plots
fig, axes = plt.subplots(2, 1, figsize=
     fizsize_with_subplots)

# Histogram of AgeFill segmented by Survived
df1 = df_train[df_train['Survived'] == 0]['Age']
df2 = df_train[df_train['Survived'] == 1]['Age']
max_age = max(df_train['AgeFill'])
axes[0].hist([df1, df2],
             bins=max_age / bin_size,
             range=(1, max_age),
             stacked=True)
axes[0].legend(('Died', 'Survived'), loc='best')
axes[0].set_title('Survivors by Age Groups Histogram')
axes[0].set_xlabel('Age')
axes[0].set_ylabel('Count')
```

Feature Engineering



Feature Engineering

Plot a normalized cross tab for AgeFill and Survived:

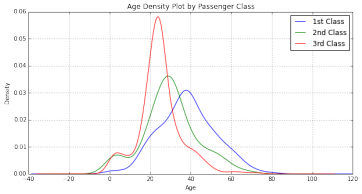


Unfortunately, the graphs above do not seem to clearly show any insights.

```
axes[1].scatter(df_train['Survived'], df_train['AgeFill'])
axes[1].set_title('Survivors by Age Plot')
axes[1].set_xlabel('Survived')
axes[1].set_ylabel('Age')
```

Feature Engineering

Plot AgeFill density by Pclass:




```
for pclass in passenger_classes:
    df_train.AgeFill[df_train.Pclass == pclass].plot(kind=
     'kde')
plt.title('Age Density Plot by Passenger Class')
plt.xlabel('Age')
plt.legend(('1st Class', '2nd Class', '3rd Class'), loc='
     best')
```

Feature Engineering

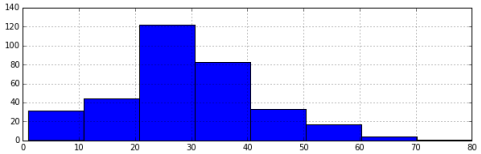
- The first class passengers were generally older then second class passengers, which in turn were older than third class passengers.
- We've determined that first class passengers had a higher survival rate than second class passengers, which in turn had a higher survival rate than third class passengers.

Feature Engineering

Plots

```
# Set up a grid of plots
fig = plt.figure(figsize=fizsize_with_subplots)
fig_dims = (3, 1)

# Plot the AgeFill histogram for Survivors
plt.subplot2grid(fig_dims, (0, 0))
survived_df = df_train[df_train['Survived'] == 1]
survived_df['AgeFill'].hist(bins=max_age / bin_size, range
     =(1, max_age))
```

Feature Engineering



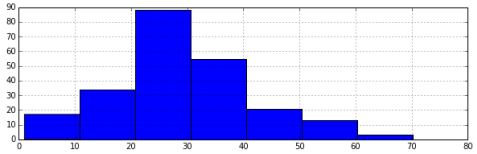
We see that most survivors come from the 20's to 30's age ranges and might be explained by the following two graphs.

Feature Engineering

Plots

```
# Plot the AgeFill histogram for Females
plt.subplot2grid(fig_dims, (1, 0))
females_df = df_train[(df_train['Sex_Val'] == 0) & (
     df_train['Survived'] == 1)]
females_df['AgeFill'].hist(bins=max_age / bin_size, range
     =(1, max_age))
```

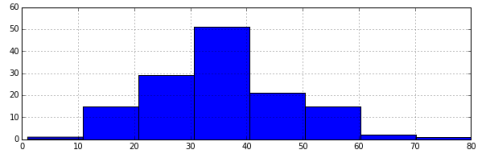
Feature Engineering



Shows most females are within their 20's.

Feature Engineering

Plots



Shows most first class passengers are within their 30's.

```
# Plot the AgeFill histogram for first class passengers
plt.subplot2grid(fig_dims, (2, 0))
class1_df = df_train[(df_train['Pclass'] == 1) & (df_train
     ['Survived'] == 1)]
class1_df['AgeFill'].hist(bins=max_age / bin_size, range
     =(1, max_age))
```

Feature Engineering

Define a new feature FamilySize that is the sum of Parch (number of parents or children on board) and SibSp (number of siblings or spouses):

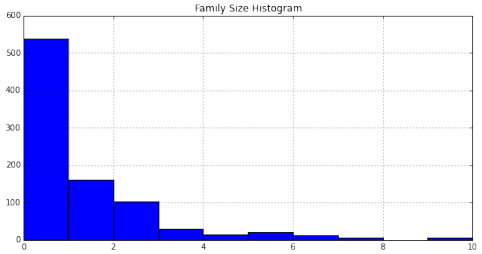
```
df_train['FamilySize'] = df_train['SibSp'] + df_train['
     Parch']
df_train.head()
```

Feature Engineering

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	Sex_Val	
0	1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171	7.2500	NaN	S	1	
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38	1	0	PC 17599	71.2833	C85	C	0	
2	3	1	3	Heikkinen, Miss. Laina	female	26	0	0	STON/O2. 3101282	7.9250	NaN	S	0	
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803	53.1000	C123	S	0	
4	5	0	3	Allen, Mr. William Henry	male	35	0	0	373450	8.0500	NaN	S	1	

Feature Engineering

Plot a histogram of FamilySize:



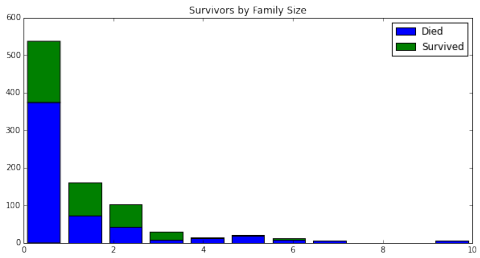

```
df_train['FamilySize'].hist()
plt.title('Family Size Histogram')
```

Feature Engineering

Plot a histogram of AgeFill segmented by Survived:

```
family_sizes = sorted(df_train['FamilySize'].unique())
family_size_max = max(family_sizes)
df1 = df_train[df_train['Survived'] == 0]['FamilySize']
df2 = df_train[df_train['Survived'] == 1]['FamilySize']
plt.hist([df1, df2],
         bins=family_size_max + 1,
         range=(0, family_size_max),
         stacked=True)
plt.legend(('Died', 'Survived'), loc='best')
plt.title('Survivors by Family Size')
```

Feature Engineering



It is not immediately obvious what impact FamilySize has on survival.

Final Data Preparation for Machine Learning

Many machine learning algorithms do not work on strings and they usually require the data to be in an array, not a DataFrame. Show only the columns of type 'object' (strings):

```
df_train.dtypes[df_train.dtypes.map(lambda x: x == 'object
    ')]

Name        object
Sex         object
Ticket      object
Cabin       object
Embarked    object
dtype: object
```

Final Data Preparation for Machine Learning

Drop the columns we won't use:

```
df_train = df_train.drop(['Name', 'Sex', 'Ticket', 'Cabin'
    , 'Embarked'],
                         axis=1)
```

Final Data Preparation for Machine Learning

Drop the columns we won't use:

- The Age column since we will be using the AgeFill column instead.
- The SibSp and Parch columns since we will be using FamilySize instead.
- The PassengerId column since it won't be used as a feature.
- The Embarked_Val as we decided to use dummy variables instead.

```
df_train = df_train.drop(['Age', 'SibSp', 'Parch', '
    PassengerId', 'Embarked_Val'], axis=1)
```

Final Data Preparation for Machine Learning

Convert the DataFrame to a numpy array:

```
train_data = df_train.values
train_data

array([[  0.    ,   3.    ,   7.25  , ...,   1.    ,  22.
        ,   1.    ],
       [  1.    ,   1.    ,  71.2833, ...,   0.    ,  38.
        ,   1.    ],
       [  1.    ,   3.    ,   7.925 , ...,   1.    ,  26.
        ,   0.    ],
       ...,
       [  0.    ,   3.    ,  23.45  , ...,   1.    ,  21.5
        ,   3.    ],
       [  1.    ,   1.    ,  30.    , ...,   0.    ,  26.
        ,   0.    ],
       [  0.    ,   3.    ,   7.75  , ...,   0.    ,  32.
        ,   0.    ]])
```

Modeling

Random Forest: Training

Create the random forest object:

```
from sklearn.ensemble import RandomForestClassifier

clf = RandomForestClassifier(n_estimators=100)
```

Random Forest: Training

Fit the training data and create the decision trees:

```
# Training data features, skip the first column 'Survived'
train_features = train_data[:, 1:]

# 'Survived' column values
train_target = train_data[:, 0]

# Fit the model to our training data
clf = clf.fit(train_features, train_target)
score = clf.score(train_features, train_target)
"Mean accuracy of Random Forest: {0}".format(score)

'Mean accuracy of Random Forest: 0.980920314254'
```

Predicting

Random Forest: Predicting

Read the test data:

	PassengerId	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	892	3	Kelly, Mr. James	male	34.5	0	0	330911	7.8292	NaN	Q
1	893	3	Wilkes, Mrs. James (Ellen Needs)	female	47.0	1	0	363272	7.0000	NaN	S
2	894	2	Myles, Mr. Thomas Francis	male	62.0	0	0	240276	9.6875	NaN	Q
3	895	3	Wirz, Mr. Albert	male	27.0	0	0	315154	8.6625	NaN	S
4	896	3	Hirvonen, Mrs. Alexander (Helga E Lindqvist)	female	22.0	1	1	3101298	12.2875	NaN	S

Note the test data does not contain the column 'Survived', we'll use our trained model to predict these values.

```
df_test = pd.read_csv('../data/titanic/test.csv')
df_test.head()
```

Random Forest: Predicting

Clean the test data similar to training data:

```
# Data wrangle the test set and convert it to a numpy
    array
df_test = clean_data(df_test, drop_passenger_id=False)
test_data = df_test.values

# Get the test data features, skipping the first column '
    PassengerId'
test_x = test_data[:, 1:]

# Predict the Survival values for the test data
test_y = clf.predict(test_x)
```

Kaggle Submission

Random Forest: Prepare for Kaggle Submission

Create a DataFrame by combining the index from the test data with the output of predictions, then write the results to the output:

```
df_test['Survived'] = test_y
df_test[['PassengerId', 'Survived']] \
    .to_csv('../data/titanic/results-rf.csv', index=False)
```

Evaluate Model Accuracy

Get an idea of accuracy before submitting to Kaggle. We'll split our training data, 80% will go to "train" and 20% will go to "test":

```
from sklearn import metrics
from sklearn.model_selection import train_test_split

train_x, test_x, train_y, test_y = train_test_split(
     train_features,

     train_target,

     test_size=0.20,

     random_state=0)
print (train_features.shape, train_target.shape)
print (train_x.shape, train_y.shape)
print (test_x.shape, test_y.shape)
((891, 8), (891,))
((712, 8), (712,))
((179, 8), (179,))
```

Evaluate Model Accuracy

Use the new training data to fit the model, predict, and get the accuracy score:

```
clf = clf.fit(train_x, train_y)
predict_y = clf.predict(test_x)

from sklearn.metrics import accuracy_score
print ("Accuracy = %.2f" % (accuracy_score(test_y,
     predict_y)))

Accuracy = 0.83
```

Evaluate Model Accuracy

View the Confusion Matrix:

```
confusion_matrix = metrics.confusion_matrix(test_y,
     predict_y)


print ("          Predicted")
print ("         |  0  |  1  |")
print ("         |-----|-----|")
print ("       0 | %3d | %3d |" % (confusion_matrix[0, 0],
                                   confusion_matrix[0, 1])
     )
print ("Actual   |-----|-----|")
print ("       1 | %3d | %3d |" % (confusion_matrix[1, 0],
                                   confusion_matrix[1, 1])
     )
print ("         |-----|-----|")
```

Evaluate Model Accuracy

Display the classification report:

```
from sklearn.metrics import classification_report
print(classification_report(test_y,
                            predict_y,
                            target_names=['Not Survived',
     'Survived']))

             precision    recall  f1-score   support
Not Survived       0.84      0.89      0.86       110
    Survived       0.81      0.72      0.76        69
 avg / total       0.83      0.83      0.82       179
```

Next?

To improve accuracy

Feature Preparation, Selection, and Engineering

- How to determine which features in your model are the most-relevant to your predictions
- Ways to reduce the number of features used to train your model and avoid over-fitting
- Techniques to create new features to improve the accuracy of your model

To improve accuracy

Model Selection and Tuning

- How other classification algorithms work
- About hyper-parameters, and how to select the hyper-parameters that give the best predictions
- How to compare different algorithms to improve the accuracy of your predictions